Teknik Antena dan Propagasi

A. Definisi Antena

1. Webster (Orang yang membuat Kamus dan Ensiklopedia)

   "*a usually metallic device (such as a rod or wire) for radiating or receiving radio waves"*

   Artinya : Sebuah perangkat yang biasanya berupa logam (seperti batang atau kawat) untuk memancarkan atau menerima gelombang radio.

2. IEEE 1975

   "*A means for radiating or receiving radio waves."*

   Artinya : Sebuah alat untuk memancarkan dan/atau menerima gelombang radio.

Modulasi adalah proses menumpangkan sinyal informasi yang berfrekuensi rendah (misal suara manusia paling besar frekuensinya 4kHz) ke sinyal carrier yang berfrekuensi jauh lebih besar daripada sinyal informasinya untuk mentransmisikan sinyal informasi agar sinyal dapat diterima dengan baik meskipun telah melewati jarak yang jauh.

B. Fungsi Antena

1. Melakukan Matching Device

   Menyesuaikan sifat-sifat atau karakteristik saluran transmisi dengan impedansi ruang bebas.

2. Sebagai Directional Device (Alat untuk mengarahkan sinyal)

   Antenna coverage dilihat dari atas (antenna ke antenna).



Omnidirectional



Sectoral

Arah pancaran sinyal salah satunya dipengaruhi oleh bentuk antenna.

- Omnidirectional → Menyebarkan sinyal ke semua arah sama besar.
- Uni directional → Satu arah, tajam seperti pemukul kasti
- Sectoral → berbentuk lidah
- Antena Yagi → untuk Televisi
- Komunikasi telestial → Point to Point.

Antenna array atau antenna susun digunakan untuk mendapatkan gain yang lebih besar agar cakupannya menjadi lebih luas, namun dengan biaya yang lebih murah. Jika cakupan masih kurang, maka bisa dilakukan penambahan daya pancar. Namun, penambahan daya pancar memiliki resiko distorsi yang lebih besar dan perlu menambah biaya lagi (mahal).

C. Parameter Antena

1. Teknik penyesuai terkait VSWR, Return Loss, dan Bandwidth
   - VSWR dan RL merupakan teknik matching.
   - VSWR yang baik untuk antena adalah ≤ 2 (paling baik 1).
   - Bandwidth di batasi oleh VSWR & RL.
2. Perangkat pengarah

   Parameter perangkat pengarah :
   - Pola radiasi : 3D, E-Plane, H-Plane
   - Intensitas Radiasi
   - Directivity
   - Gain
   - Polarisasi
3. Impedansi :
   - Impedansi Diri (Self Impedance)
   - Impedansi Gandeng (Mutual Impedance)
4. Friis Transmission Formula

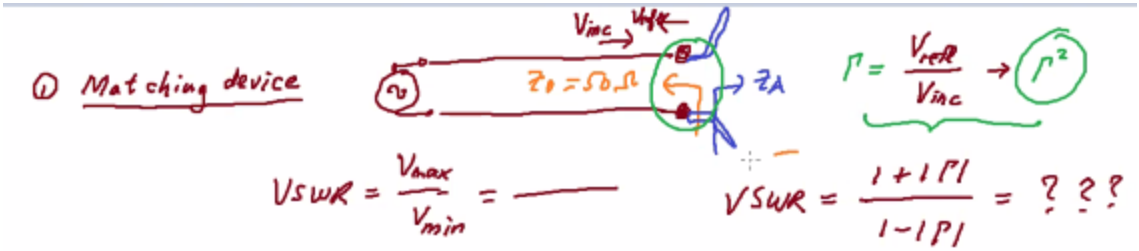
Hal-hal yang Akan Dipelajari

- Antena Dipol
- Antena Mikrostrip
- Antena Susun (Array)
- Antena Reflektor → antenna yang memanfaatkan pantulan untuk melipatgandakan gain (teorema bayangan)
- Beberapa Jenis Antena
- Konsep Teknik Pengukuran Antena

Fungsi Antena

1. Matching Device

Menyesuaikan impedansi karakteristik saluran transmisi dengan impedansi di ruang bebas.



Zo = 50 Ω → impedansi saluran

ZA → impedansi antena

Impedansi antena harus menyesuaikan dengan impedansi saluran 50 ohm

Zo dengan ZA berhadapan, yang akan dihubungkan.

$V_{incident}$ mau diberikan ke antena seluruhnya, tetapi kenyataannya ada yang dikembalikan sebesar $V_{refleksi}$.

$V_{refleksi}$ yang diinginkan tentunya sekecil mungkin.

Faktor Pengukuran Matching Device

a) Faktor Refleksi (τ) idealnya 0.

$$\tau = \frac{V_{refleksi}}{V_{incident}}$$

Daya (P) = τ^2

b) VSWR (*Voltage Standing Wave Ratio* / Rasio Tegangan Gelombang Berdiri)

$$VSWR = \frac{V_{maks}}{V_{min}}$$

$$VSWR = \frac{1 + |\tau|}{1 - |\tau|}$$

c) Return Loss (RL) → Ukuran faktor refleksi dalam skala logaritmis

$$RL_{[dB]} = -20 \log |\tau|$$

Harus diubah dalam skala logaritmis agar dapat menampilkan sinyal dengan range yang lebar.

-90 dbm = $1/10^9$ mW

-30 dB = $1/10^3$ W

Misal impedansi karakteristik saluran transmisi Zo = 50 Ω , dan impedansi antena ZA = 70 Ω

$$\tau = \frac{Z_A - Z_o}{Z_A + Z_o} = \frac{70 - 50}{70 + 50} = \frac{1}{6}$$

$$RL_{[dB]} = -20 \log \left|\frac{1}{6}\right| = 15.56dB = 16\ dB$$

15.56 dB = $10^{15,56/10}$ = $10^{1,556}$

= 35,9 = 36

RL sebesar 36 artinya ada daya yang dikembalikan sebesar 1/36 dari daya yang masuk.

Misal daya yang diberikan 100mW, maka 1/36 x 100mW = 2,8. Artinya ada daya yang dikembalikkan sebesar 2,8 mW.

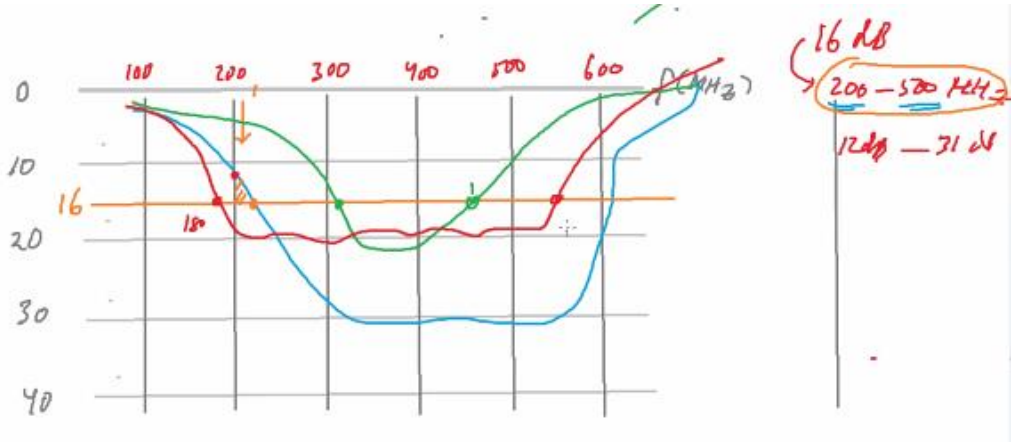
RL untuk microstrip yang paling buruk adalah 10 dB. Ada Daya yang dikembalikan 10%

RL = 10 dB = $10^{10/10}$ = 10

→ $\frac{1}{10}$ x 100% → 10% dipantulkan, 90 % diserap

RL = 16 dB = $10^{16/10}$ = 39.81

→ $\frac{1}{39.81}$ x 100% → 2,5% dipantulkan, 97,5% diserap



2. Directional Device (Pengarah)

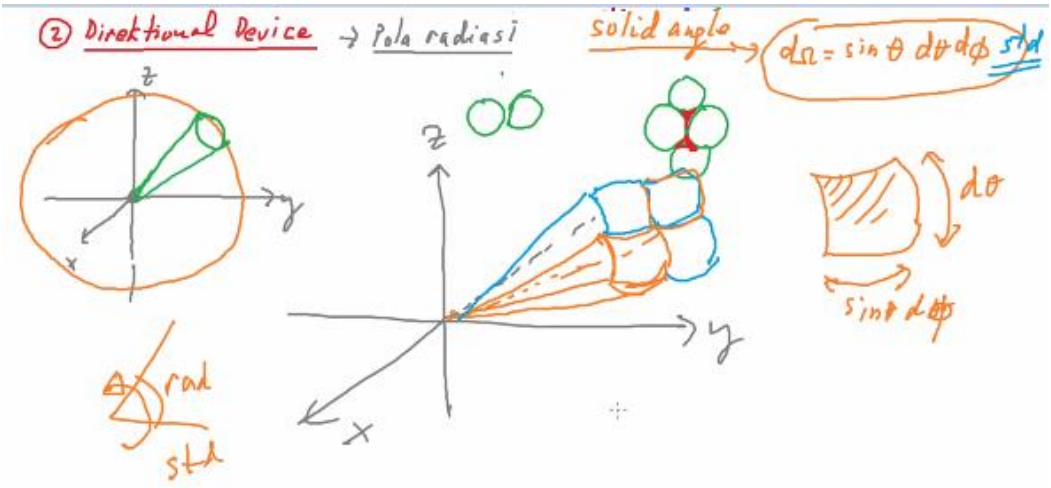
Antenna pada hakikatnya bisa merespons dan mengarahkan.

Macam-macam pola radiasi:

- Unidirectional → ke satu arah (polanya tajam seperti tongkat baseball)
- Sectoral → mengarahkan ke satu sector saja seperti lidah
- Omnidirectional → mengarahkan sinyal ke semua arah dan sama besar (lingkaran)

Intensitas radiasi adalah daya yang diradiasikan oleh antena per sudut (dibayangkan seperti kerucut).

Volume Garis = 0.

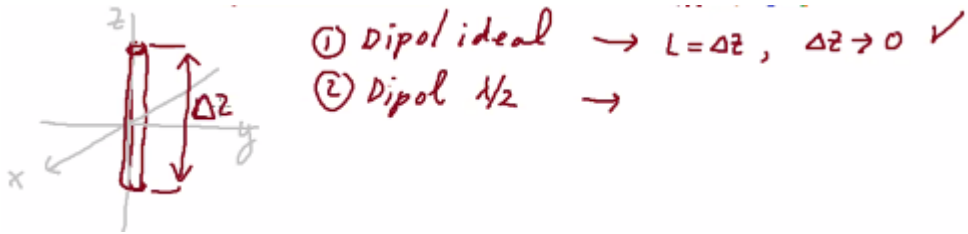


$d\theta$ = 0 sampai 180◦

$d\Omega = \sin\theta . d\theta . d\emptyset$

Elemen yang digambarkan disebut *solid angle* (sudut padat)

Antena Dipol



Macam-macam antena dipol, diantaranya : antena dipol ideal dan antena ½ lambda.

1. Antena dipol ideal → suatu dipol yang panjangnya L = ΔZ, ΔZ mendekati 0. Namun tidak aplikatif.

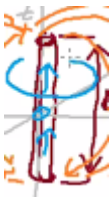
   Berbicara dipol ideal → persamaan maxwell.

   Panjangnya L = ΔZ dan diletakkan di sumbu Z.

   Sifat dipol ideal, yaitu adanya pergerakan medan listrik dari positif ke negatif.



Warna Orange → Medan Listrik



Kemudian disekelilingnya terdapat medan magnet.

Perhitungan Pada Dipol Ideal

- Medan Listrik

$$\vec{E} = \frac{I\,\Delta Z}{4\pi}\, j\omega\mu\, \frac{e^{-j\beta r}}{r}\, sin\theta\, \hat{a}_\theta$$

- Medan Magnet

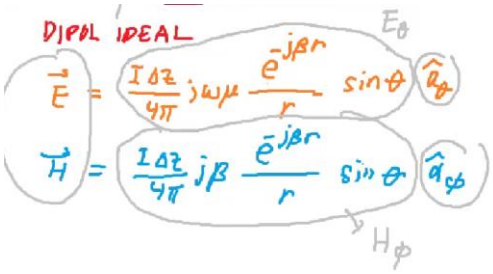
$$\vec{H} = \frac{I\,\Delta Z}{4\pi}\, j\beta\, \frac{e^{-j\beta r}}{r}\, sin\theta\, \hat{a}_\phi$$



Kombinasi antara medan listrik dengan medan magnet arahnya ke $\vec{E}\ x\ \vec{H}$ yang artinya putarlah sekrup dari arah E ke H.



Memutar sekrup ke kiri, maka mengeluarkan sekrup, sehingga arah daya menuju kita.



H mengarah ke Azimuth → 0 – 180°

E mengarah ke Elevasi → 0 - 360°

E_θ komponen yang mengarah ke θ , yaitu dari kutub utara ke kutub selatan.

$$\frac{E_\theta}{H_\emptyset} = \frac{\omega\,\mu}{\beta} = \frac{\omega\,\mu}{\omega\sqrt{\mu\,E}} = \sqrt{\frac{\mu}{E}} = \eta$$

Untuk ruang bebas :

$\eta_o = 120\,\pi = 376.7\,\Omega$ [impedansi intrinsik]

Daya Untuk Antena Dipol Ideal

$$P = \frac{1}{2}\iint \vec{E}\ x\ \vec{H}\,.d\vec{S}$$

$$P = \frac{\omega\,\mu\,\beta}{12\pi}(I\,\Delta Z)^2$$

Parameter Dasar Antena Dipol Ideal

1) Pola Elemen Antena

   g(θ, $\emptyset$) = Sin θ

   $0 \leq\ \theta \leq\ \pi$

   $0 \leq\ \emptyset \leq\ 2\pi$

2) Directivity

   D = 1,5 = 1,77 dB

   Gain

   G = e . D ; e = efisiensi antena, efisiensi radiasi, return loss

3) Impedansi

   $\boldsymbol{Z = 800\left(\frac{\Delta Z}{\lambda}\right)\ \Omega}$

   $\Delta Z \leq\ 0.1\,\lambda$

   $\Delta Z \leq\ 0.01\,\lambda$

4) Apertur Efektif

$\boldsymbol{A_e = 0.119\ \lambda^2}$

Misal f = 100 MHz

$\lambda = \frac{c}{f} = \frac{300\ x\ 10^6}{100\ x\ 10^6} = 3m = 300\ cm$

$A_e = 0.119\ (300)^2$ = 10710 cm^2 = 1,071 m^2

5) HPBW (Half Power Bandwidth)

- **HPBW-E → HP 1 = 90°**
- **HPBW-H → HP 2 = 360°**

2. Antena dipol ½ lambda $\left(\frac{\lambda}{2}\right)$ → mempelajari penurunan sifat array, coupling, dan sifat imbas. Panjang L = $\frac{\lambda}{2}$

Parameter Dasar Antena Dipol ½ lambda

1) Pola Elemen Antena

$\mathbf{g}(\boldsymbol{\theta}, \emptyset) = \frac{\boldsymbol{Cos\ \left(\frac{\pi}{2} Cos\theta\right)}}{\boldsymbol{Sin\ \theta}}$

$0 \leq\ \theta \leq\ \pi$

$0 \leq\ \emptyset \leq\ 2\pi$

2) Directivity

D = 1,64 = 2,15 dB

Gain

G = e . D ; e = efisiensi antena, efisiensi radiasi, return loss

3) Impedansi

$\boldsymbol{Z = 73 + j\ 42.5\ \Omega}$

$\Delta Z \leq\ 0.1\ \lambda$

$\Delta Z \leq\ 0.01\ \lambda$

4) Apertur Efektif

$\boldsymbol{A_e = 0.13\ \lambda^2}$

Misal f = 100 MHz

$\lambda = \frac{c}{f} = \frac{300\ x\ 10^6}{100\ x\ 10^6} = 3m = 300\ cm$

$A_e = 0.13\ (300)^2$ = 11700 cm^2 = 1,17 m^2

5) HPBW (Half Power Bandwidth)
   - **HPBW-E → HP 1 = 78°**
   - **HPBW-H → HP 2 = 360°**

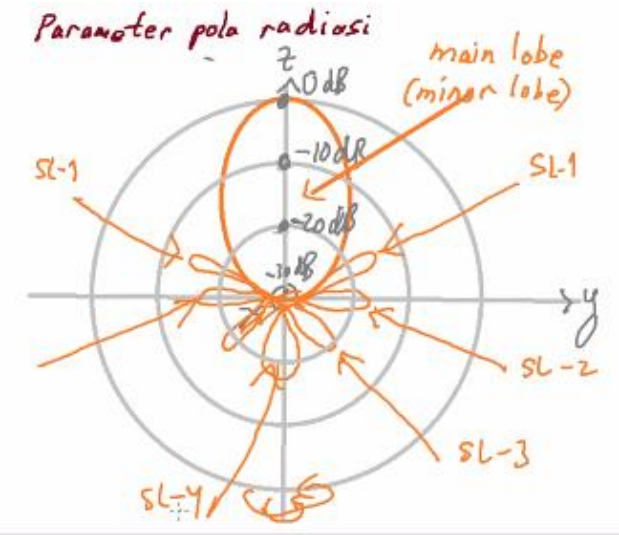
Tabel Rangkuman Parameter Antena

Parameter Antena	Dipol Ideal	Dipol ½ lambda	Isotropis
Pola Elemen g$(\theta, \emptyset)$	Sin θ $0 \leq \theta \leq \pi$ $0 \leq \emptyset \leq 2\pi$	$\frac{Cos\left(\frac{\pi}{2}Cos\theta\right)}{Sin\,\theta}$ $0 \leq \theta \leq \pi$ $0 \leq \emptyset \leq 2\pi$	1 $0 \leq \theta \leq \pi$ $0 \leq \emptyset \leq 2\pi$
Impedansi (Z)	$Z = 800\left(\frac{\Delta Z}{\lambda}\right)\ \Omega$	$Z = 73 + j\,42.5\ \Omega$	$120\pi = 376.7\ \Omega$
Directivity (D)	1,5 atau 1,77 dB	1,64 atau 2,15 dB	1 atau 0 dB
Aperture Efektif (A_e)	$0.119\ \lambda^2$	$0.13\ \lambda^2$	?
Half Power Bandwidth (HPBW)	• HPBW-E = 90° • HPBW-H = 360°	• HPBW-E = 78° • HPBW-H = 360°	• HPBW-E = 360° • HPBW-H = 360°

Parameter Pola Radiasi

Kita hanya bisa melihat 2 sisi bidang orthogonal, dilihat dari manapun bola akan tetap terlihat sama.





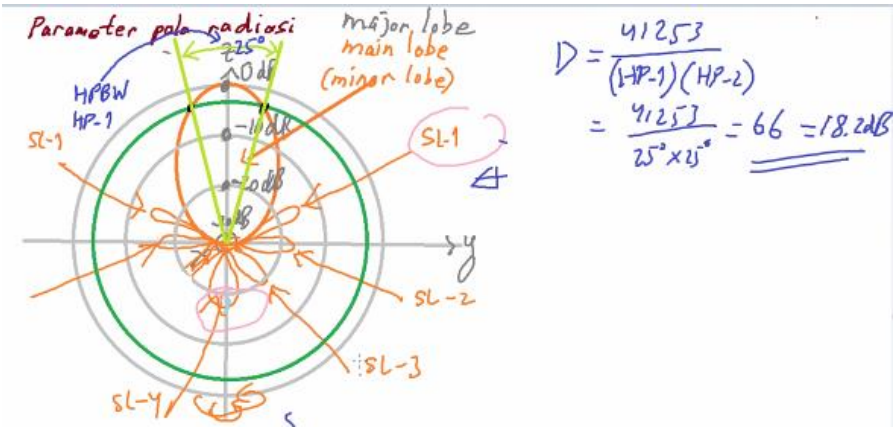
- **Mine Lobe / Minor Lobe / Major Lobe** adalah lobe paling besar yang berisi sebagian besar dari daya gelombang elektromagnetik yang dipancarkan dan menunjukkan arah (menuju positif)
- Lobe yang kecil-kecil disebut Side Lobe. Jika diputar, Side Lobenya nyambung.
- SL-4 itu berlawanan sehingga disebut oposisi. Nanti muncul perhitungan besarnya penekanan back lobe.
- Side Lobe yang diukur hanya SL-1 dan SL-4 (back lobe). Sl-2, SL-3 tidak dihitung karena sangat kecil.

  Jika melihat gambar, maka

  Side Lobe Level (SLL) 1 = -16 dB

  Back Lobe Level (BLL) = Front to Side Ratio (FBR)

  FBR = -18 dB
- Untuk menemtukan HPBW, maka turun 3dB



Karena simetris, maka HP-1 = HP-2. Misal HP-1 = 25°

- **Pendekatan Directivity**

$$\boldsymbol{D = \frac{41253}{(HP-1)(HP-2)}}$$

$$D = \frac{41253}{(25°)(25°)} = \frac{41253}{625}$$

D = 66

D [dB] = 10 log 66 = 18.2 dB

- **Perhitungan Murni Directivity (D)**

Directivity → Perbandingan antara intensitas radiasi maksimum dibagi dengan intensitas radiasi rata-rata.

$$\boldsymbol{D = \frac{U_m}{U_{ave}}}$$

$$\boldsymbol{D = \frac{4\pi}{\Omega_A}} \qquad ;\Omega = \text{omega}$$

Directivity Antena Dipol Ideal

$$\boldsymbol{\Omega_A = \int_0^{2\pi}\int_0^{\pi} (\, g^2(\theta,\emptyset)\,)\, sin\theta\, d\theta\, d\emptyset} \quad ;\ g(\theta,\emptyset) = sin\theta$$

$$\Omega_A = \int_0^{2\pi} d\emptyset \int_0^{\pi} (\, sin^2\theta)\, sin\theta\, d\theta$$

$$\int_0^{2\pi} d\emptyset = 2\pi$$

$$\int_0^{\pi} (\, 1 - cos^2\theta)\, dcos\theta$$

$$-\int_0^{\pi} dcos\theta + \int_0^{\pi} cos^2\theta\, dcos\theta$$

$$= -cos\theta\,|_0^{\pi} + \frac{1}{3}cos^3\theta\,|_0^{\pi} \quad ;\cos\pi = -1\,, \cos 0 = 1$$

$$= -(-1-1) + \frac{1}{3}(-1-1)$$

$$= 2 - \frac{2}{3} = \frac{6}{3} - \frac{2}{3} = \frac{4}{3}$$

$$\Omega_A = 2\pi\, x\, \frac{4}{3} = \frac{8\pi}{3}$$

$$D = \frac{4\pi}{\Omega_A} = \frac{4\pi}{1}\ x\ \frac{3}{8\pi}$$

D = 1.5

- **Efisiensi Antena Dipol (e)**

$$e = \frac{P_r}{P_{in}} = \frac{P_r}{P_r + \rho_{ohmic}}$$

P_r = Daya yang diradiasikan

P_{in} = Daya yang diterima oleh antena

ρ_{ohmic} harusnya 0, tapi tidak mungkin.

$$e = \frac{R_{ri}}{R_{ri} + R_{ohmic}}$$

Contoh Soal :

Antena dipol ideal f = 1 MHz, ΔZ = 1m

$$R_{ohmic} \approx \frac{L}{2\pi a}\ x\ R_s$$

L = panjang wire = 1m, a = radius kawat, Rs = surface resistance

$$R_s = \sqrt{\frac{\omega\ \mu}{2\sigma}}$$

$\omega = 2\pi f = 2\pi\ x\ 10^6 = 6.28\ x\ 10^6$

$\mu_0 = 4\pi\ x\ 10^{-7}$

konduktivitas $\sigma = 5.7\ x\ 10^7\ \mho$

ΔZ = 1m = 0.0033 λ

$$\lambda = \frac{c}{f} = \frac{300\ x\ 10^6}{10^6} = 300m$$

$$R_s = \sqrt{\frac{(6.28\ x\ 10^6)\ (12.56\ x\ 10^{-7})}{2(5.7\ x\ 10^7)}}$$

$$\boldsymbol{R_s = 2.63\ x\ 10^{-4}\ \Omega}$$

$$R_{ohmic} \approx \frac{1}{2\pi(4.06\ x\ 10^{-4})}\ x\ 2.63\ x\ 10^{-4}$$

$$\boldsymbol{R_{ohmic} \approx 0.103\ \Omega}$$

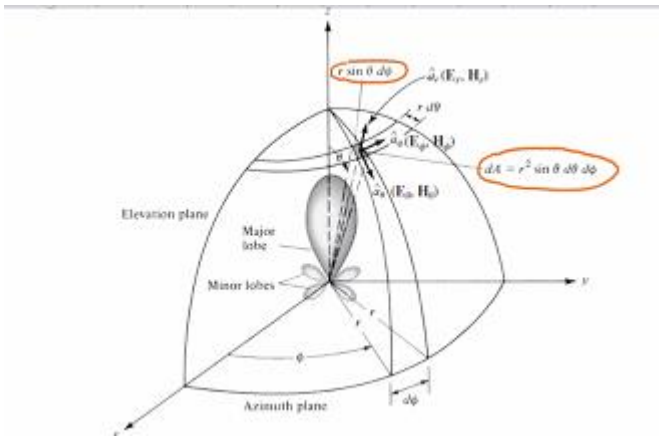
$$R_{ri} = 800\left(\frac{\Delta Z}{\lambda}\right)^2 = 800\left(\frac{1}{300}\right)^2$$

$$\boldsymbol{R_{ri} = 0.0088\,\Omega}$$

$$e = \frac{0.0088}{0.0088 + 0.103}\ x\ 100\%$$

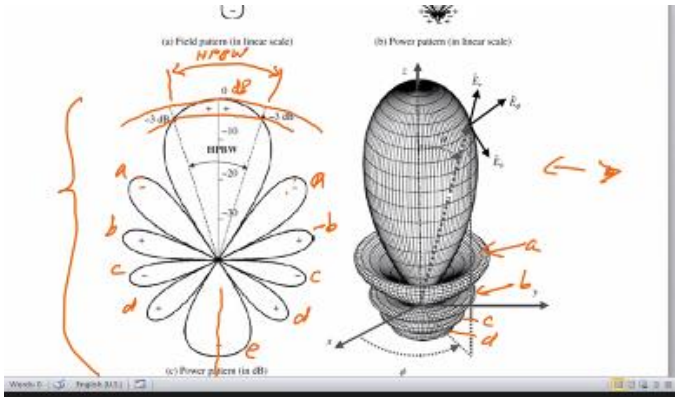
$\boldsymbol{e = 7.87\%}$ (yang dibuang sebesar 92.13%)

Pola Radiasi

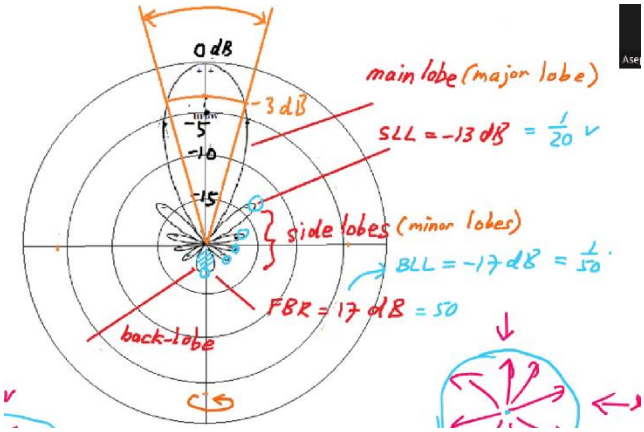


1. Koordinat kartesian A(x,y,z)
2. Koordinat silinder A($\rho, \emptyset, z$)
3. Koordinat bola A(r,$\theta, \emptyset$)

Pola Radiasi 3D



Sulit menampilkan pola radiasi 3D yang benar, maka dari itu diukur pada principle plan, yaitu pada half plane dan e-plane.



Selalu ada bocoran (side lobe).
Side lobe yang berlawanan dengan major lobe disebut backlobe.

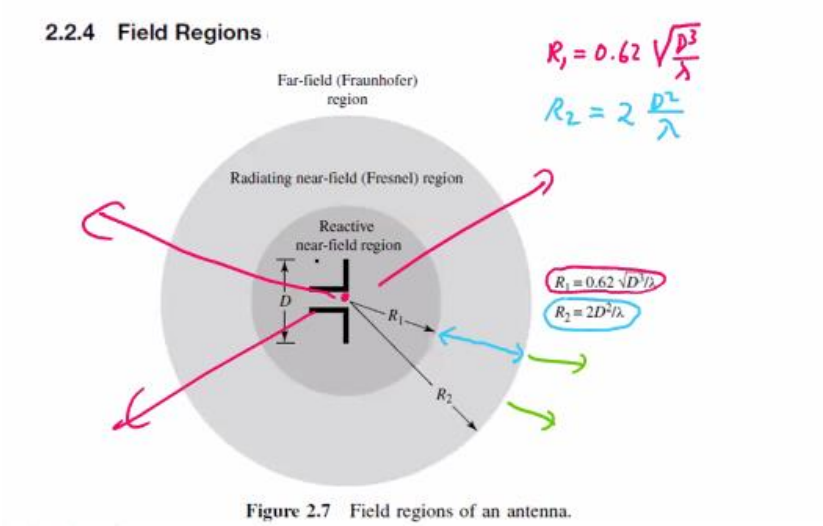
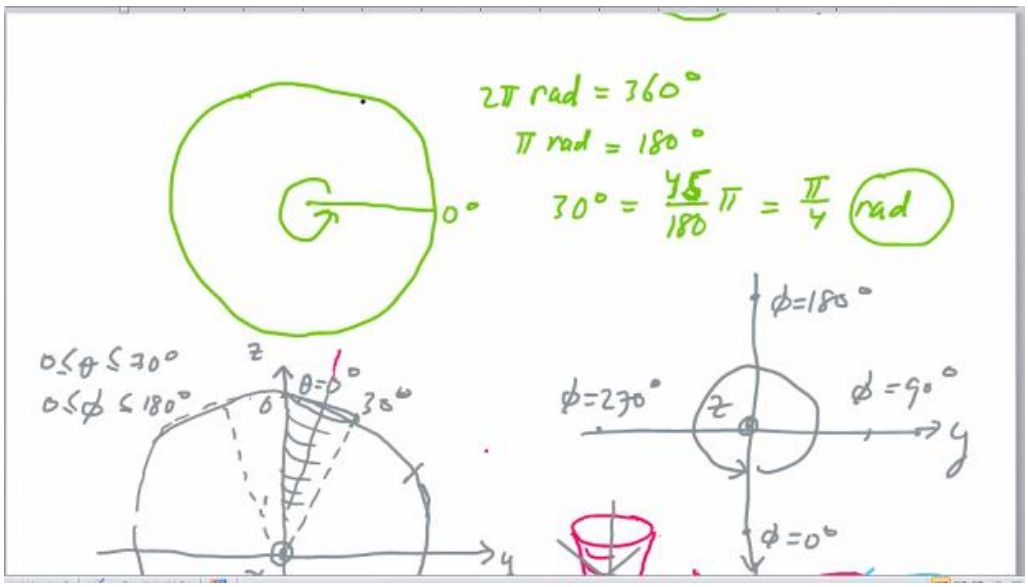
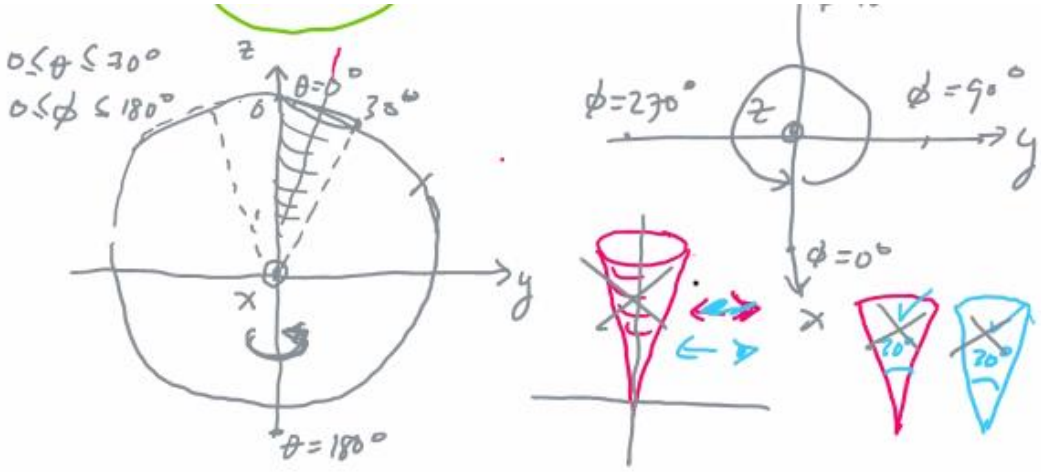


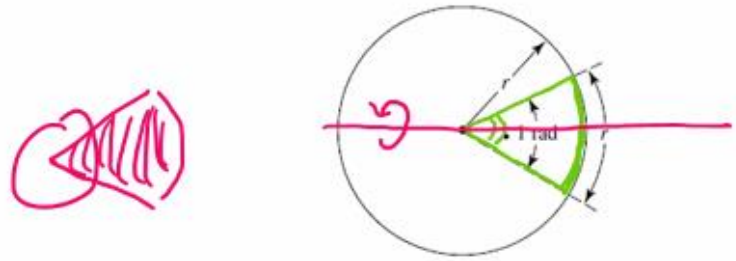
Figure 2.7 Field regions of an antenna.

$$R_1 = 0.62\sqrt{\frac{D^3}{\lambda}} \qquad R_2 = 2\frac{D^2}{\lambda}$$



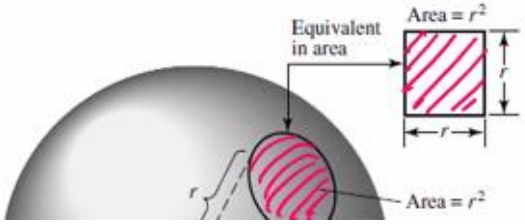


2.2.5 Radian and Steradian

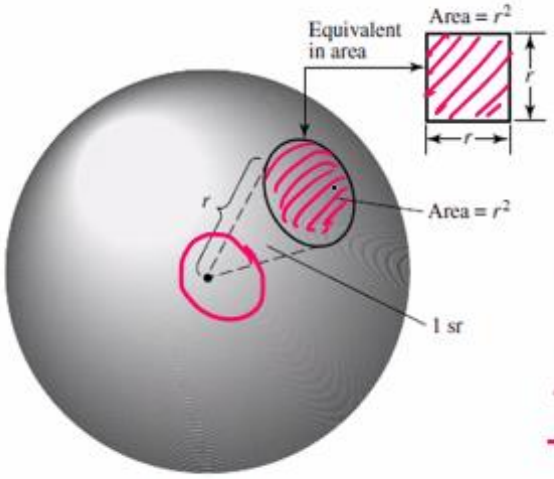


(a) Radian

cone
conelo



(a) Radian



(b) Steradian

solid angle = Ω

$A \rightarrow dA$

$\Omega \rightarrow d\Omega$

Sistem Bola

$$dA = r^2 \sin\theta \, d\theta \, d\phi \quad (\text{m}^2) \qquad (2\text{-}1)$$

$$d\Omega = \frac{dA}{r^2} = \sin\theta \, d\theta \, d\phi \quad (\text{sr}) \qquad (2\text{-}2)$$

$$\Omega_A = \int_{\phi=0}^{360^\circ} \int_{\theta=0}^{30^\circ} \sin\theta \, d\theta \, d\phi$$

,



$e = e_1 \times e_2 \rightarrow e_2 = e_c \cdot e_d$

$0.2 \times 90 = 18$

$e_2 = e_{cd}$

$10\ dB = 10 \rightarrow e_1 = 0.9$

$13\ dB = 20 \rightarrow e_1 = 0.95$

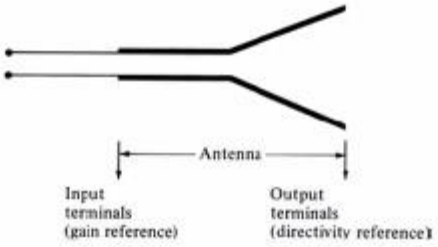
$e = 0.9 \times 0.8$

$e_2 = 0.8$ ✓

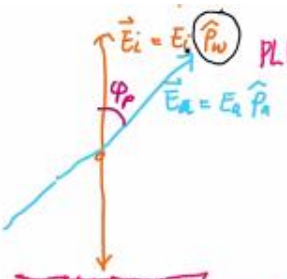
$e = 0.72$

72 watt

2.8 ANTENNA EFFICIENCY



(a) Antenna reference terminals



$PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2 = (|\hat{\rho}_w||\hat{\rho}_a| \cos\varphi_p)^2$

$= ((1)(1)\cos\varphi_p)^2$

$PLF = \cos^2\varphi_p$

$PLF = |\cos\varphi_p|^2$

2.12.2 Polarization Loss Factor and Efficiency

Assuming that the electric field of the incoming wave can be written as

$$\mathbf{E}_i = \hat{\rho}_w E_i \qquad (2\text{-}69)$$

where $\hat{\rho}_w$ is the unit vector of the wave, and the polarization of the electric field of the receiving antenna can be expressed as

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\alpha$$

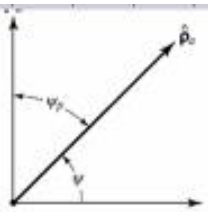
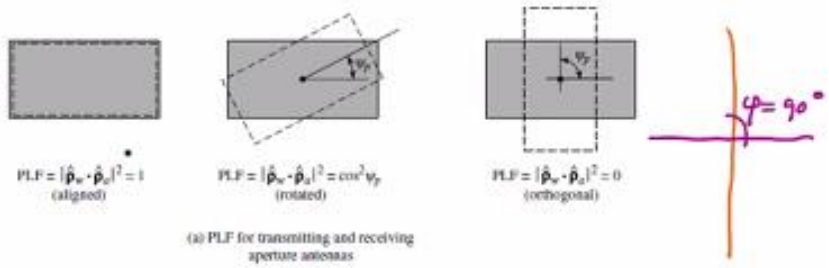


Figure 2.24 Polarization unit vectors of incident wave ($\hat{\rho}_w$) and antenna ($\hat{\rho}_a$), and polarization loss factor (PLF).



(a) PLF for transmitting and receiving aperture antennas



Example 2.11

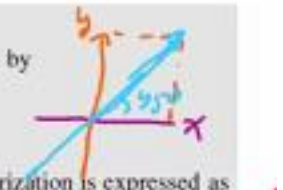
The electric field of a linearly polarized electromagnetic wave given by

$$\mathbf{E}_i = \hat{\mathbf{a}}_x E_0(x, y) e^{-j\beta z}$$

is incident upon a linearly polarized antenna whose electric-field polarization is expressed as

$$\mathbf{E}_a = (\hat{\mathbf{a}}_x + \hat{\mathbf{a}}_y) E(r, \theta, \phi)$$

Find the polarization loss factor (PLF).



$$\frac{\hat{a}_x + \hat{a}_y}{\sqrt{1+1}} = \frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}}$$

Solution: For the incident wave

$$\hat{\rho}_w = \hat{\mathbf{a}}_x$$

and for the antenna

$$\hat{\rho}_a = \frac{1}{\sqrt{2}}(\hat{\mathbf{a}}_x + \hat{\mathbf{a}}_y)$$

The PLF is then equal to

$$\text{PLF} = |\hat{\rho}_w \cdot \hat{\rho}_a|^2 = \left|\hat{\mathbf{a}}_x \cdot \frac{1}{\sqrt{2}}(\hat{\mathbf{a}}_x + \hat{\mathbf{a}}_y)\right|^2 = \frac{1}{2}$$

which in dB is equal to

$$\text{PLF (dB)} = 10\log_{10}\text{PLF (dimensionless)} = 10\log_{10}(0.5) = -3$$

$$\hat{G} = \hat{a}_x + \hat{a}_y \Rightarrow \hat{a}_G = \frac{\hat{a}_x + \hat{a}_y}{\sqrt{\hat{a}_x^2 + \hat{a}_y^2}} = \frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}}$$

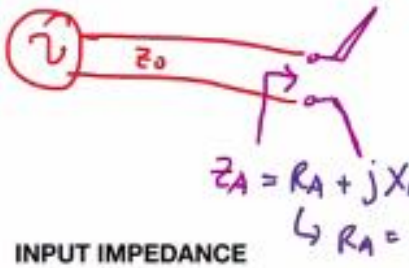
$\vec{A} \rightarrow |\vec{A}| \quad \hat{a}_A = \dfrac{\vec{A}}{|\vec{A}|}$

Example 2.11

The electric field of a linearly polarized electromagnetic wave given by

$$\mathbf{E}_i = \hat{\mathbf{a}}_x E_0(x, y) e^{-jkz}$$

is incident upon a linearly polarized antenna whose electric-field polarization is expressed as



2.13 INPUT IMPEDANCE

Input impedance is defined as "the impedance presented by an antenna at its terminals or the ratio of the voltage to current at a pair of terminals or the ratio of the appropriate components of the electric to magnetic fields at a point."

$$Z_A = R_A + jX_A \tag{2-72}$$

$$R_A = R_r + R_L \tag{2-73}$$

R_r = radiation resistance of the antenna
R_L = loss resistance of the antenna

$$P = \frac{1}{2}\iint (\mathbf{E} \times \mathbf{H}^*) \cdot d\mathbf{s} \tag{2-168}$$

$$R_r = 80\pi^2 \left(\frac{\Delta z}{\lambda}\right)^2 \Omega \qquad \textit{ideal dipole} \tag{2-169}$$

$$e = \frac{P_r}{P_{in}} = \frac{P_r}{P_r + P_{ohmic}} = e_2 \tag{1-181}$$

$$e = \frac{\frac{1}{2}R_{ri}|I_{in}|^2}{\frac{1}{2}R_{ri}|I_{in}|^2 + \frac{1}{2}R_{ohmic}|I_{in}|^2}$$

$$= \frac{R_{ri}}{R_{ri} + R_{ohmic}} = \frac{R_{ri}}{R_{in}} = \frac{R_r}{R_r + R_{ohmic}} = \frac{R_r}{R_{in}} \tag{1-182}$$

For many antennas radiation efficiency is nearly 100%. For electrically small antennas, however, the radiation efficiency may be extremely small. A quick calculation will show this. Consider an ideal dipole operating at 1 MHz and of length $\Delta z = 1$ m $= 0.0033\lambda$. It has a radiation resistance from (1-180) of

$$R_r = 80\pi^2 \left(\frac{1}{300}\right)^2 = 0.0088 \text{ ohm.} \tag{1-183}$$

$P_L = P_{ohmic}$

$e_2 = e_{cd}$

$$e = \frac{P_r}{P_{in}} = \frac{P_r}{P_r + P_{ohmic}} = e_2 \qquad (1\text{-}181)$$

$$e = \frac{\frac{1}{2}R_{ri}|I_{in}|^2}{\frac{1}{2}R_{ri}|I_{in}|^2 + \frac{1}{2}R_{ohmic}|I_{in}|^2}$$

$$= \frac{R_{ri}}{R_{ri} + R_{ohmic}} = \frac{R_{ri}}{R_{in}} = \frac{R_r}{R_r + R_{ohmic}} = \frac{R_r}{R_{in}} \qquad (1\text{-}182)$$

Contoh Soal :

1. Antena isotropik hipotetis memancar di ruang bebas. Pada jarak 100m dari antena, medan listrik totalnya adalah 5V/m. Tentukan Power Density (Wrad) dan Power Radiated (Prad)!

   Dik :

   Impedansi intrinsik ruang bebas : $\eta_o = 120\,\pi = 376.7\,\Omega$

   R = 100 m

   $E_\theta = \frac{5V}{m}$

   Jawab :

   a. Power Density (Wrad)

   $$H = \frac{E}{\eta}$$

   $$\vec{E}\ x\ \overrightarrow{H^*} = \left|\vec{E}\right|\left|\overrightarrow{H^*}\right| \sin\alpha$$

   $$= E\frac{E}{\eta} = \frac{E^2}{\eta}$$

   $$W_{rad} = \frac{1}{2}\left[\vec{E}\ x\ \overrightarrow{H^*}\right] = \frac{1}{2}\frac{E^2}{\eta}\ \hat{a}_r$$

   $$W_{rad} = \frac{5^2}{2\,(120\,\pi)} = 0.03315\ \hat{a}_r\ \ Watts/m^2$$

   b. Power Radiated (Prad)

   $$dS = r^2\ sin\theta\ d\theta\ d\emptyset$$

   $$P_{rad} = \oint_S W_{rad}\ dS = \int_{\emptyset=0}^{2\pi}\int_{\theta=0}^{\pi} W_{rad}\ r^2\ sin\theta\ d\theta\ d\emptyset$$

   $$P_{rad} = 0.03315 \int_{\emptyset=0}^{2\pi} d\emptyset \int_{\theta=0}^{\pi} 100^2\ sin\theta\ d\theta\ d\emptyset$$

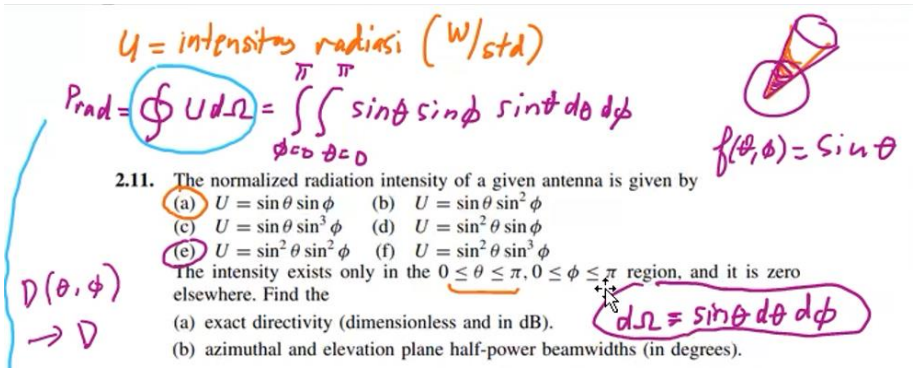
$$P_{rad} = (0.03315)(2\pi)(100^2)[-cos\theta\ |_0^{\pi}]$$

$$P_{rad} = (0.03315)(2\pi)(100^2)[-(-1-1)]$$

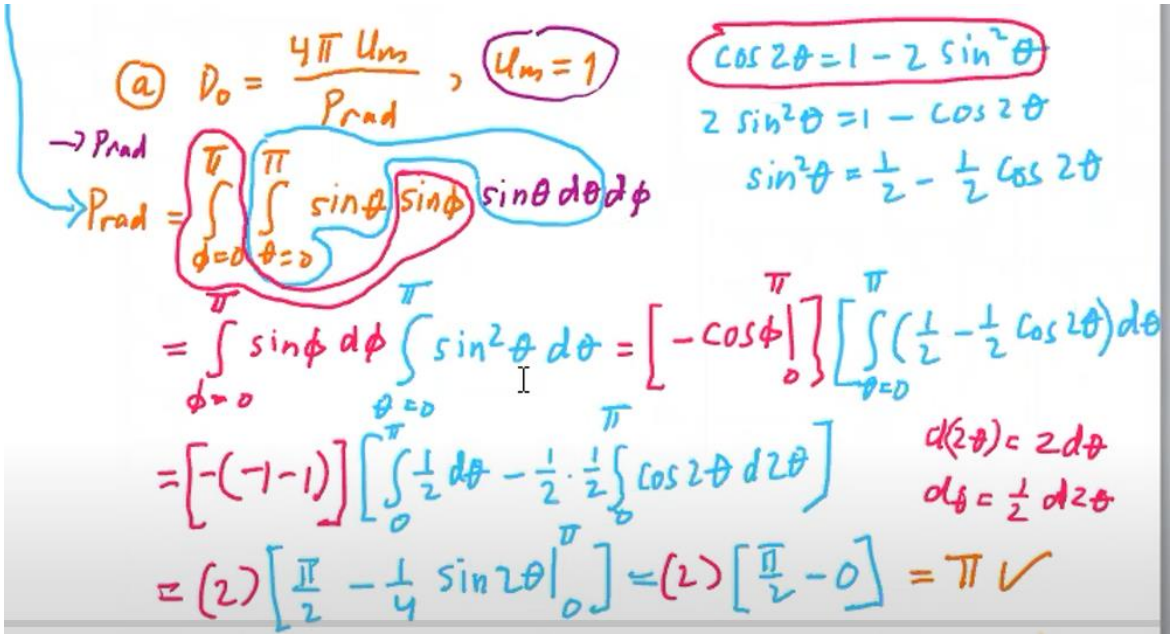
$$P_{rad} = (0.03315)(2\pi)(100^2)(2)$$

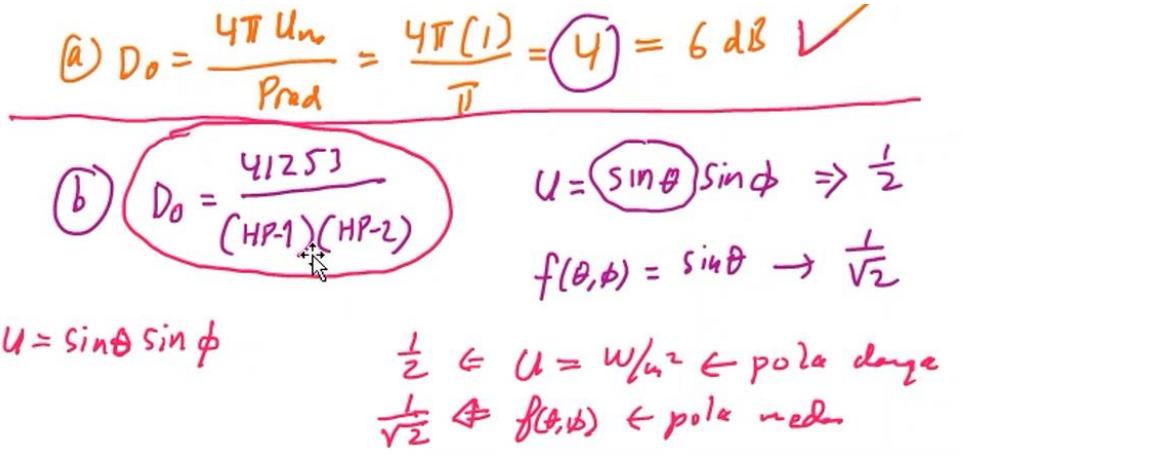
$$P_{rad} = 4165.75\ Watts$$

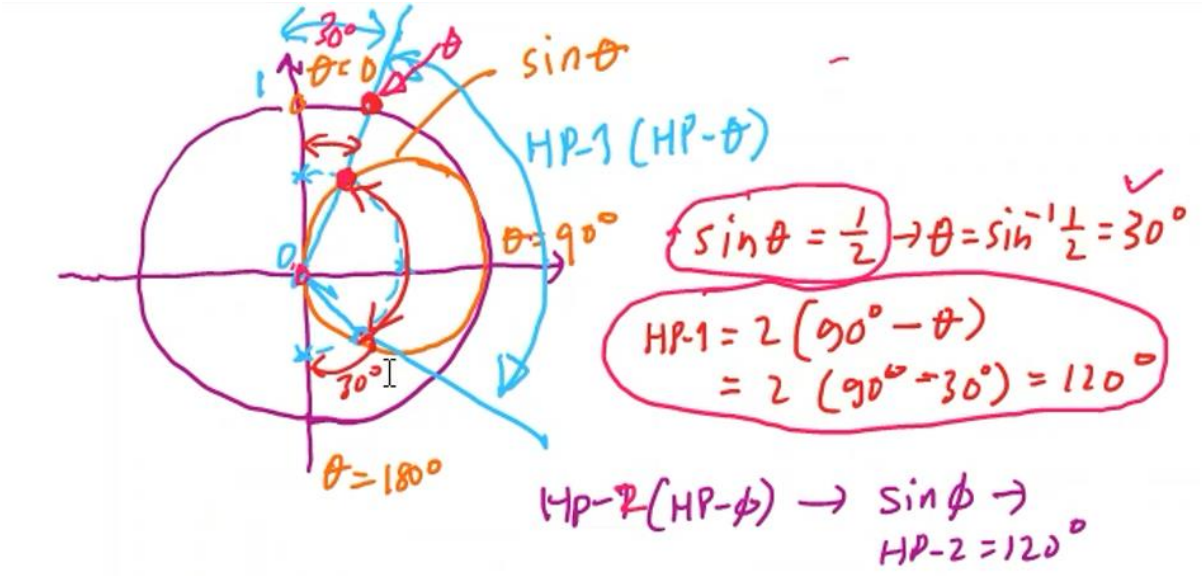
2. Soal



Jawab :

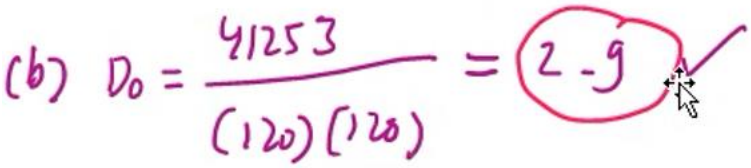


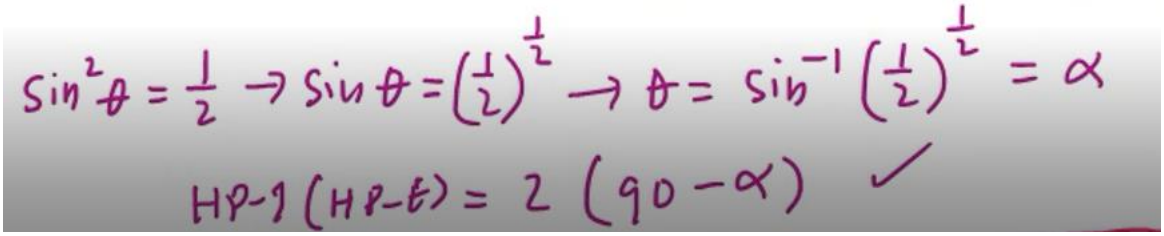




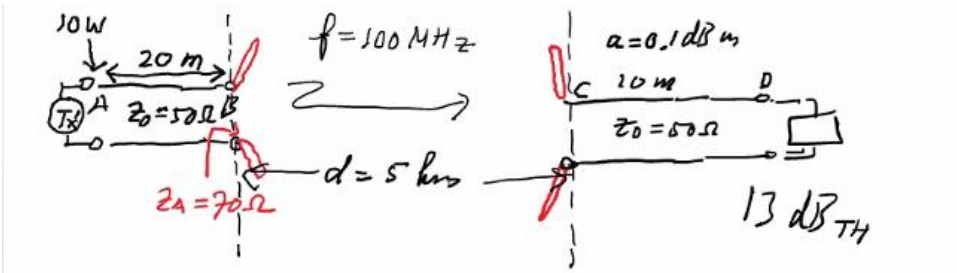
HP-1 dibidang teta (Azimuth) = 120 derajat, HP-2 di bidang phi (elevasi) = 120 derajat.

Cara pendekatan :





Daya dan Pantulan



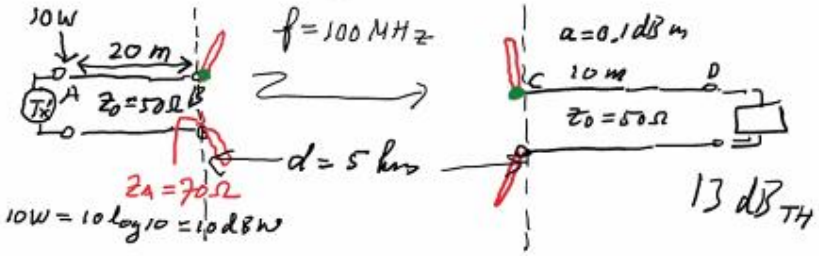
$P_A = 10W = 30\ dBm$, $a_T = 3\ dB \rightarrow P_B = 30\ dBm - 3dB = 27\ dBm$

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{70-50}{70+50} = \frac{1}{6} \rightarrow RL = -20 \log \frac{1}{6} = 16\ dB$$

DIPOL

$$R_{in} = 30\left[\left(1 - \cot^2 \frac{\beta L}{2}\right) \text{Cin } 2\beta L + 4 \cot^2 \frac{\beta L}{2} \text{Cin } \beta L + 2 \cot \frac{\beta L}{2} (\text{Si } 2\beta L - 2\,\text{Si } \beta L)\right]$$

Redaman kabel = 0.5 dB/m



$10W = 10 \log 10 = 10\ dBW$

$P_A = 10W = 40\ dBm$, $a_T = 3\ dB \rightarrow P_B = 40\ dBm - 3dB = 37\ dBm$

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{70-50}{70+50} = \frac{1}{6} \rightarrow RL = -20 \log \frac{1}{6} = 16\ dB$$

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{70-50}{70+50} = \frac{1}{6} \rightarrow RL = -20 \log \frac{1}{6} = 16\ dB$$

$$P_T = 5 - \frac{1}{40}(5) = \frac{39}{40} \times 5\ W = 4.875\ W = 6,88\ dBW$$

$$P_T = 36,88\ dBm$$

FRIIS TRANSMISSION FORMULA

$$P_R(dBm) = P_T(dBm) + G_T(dB) + G_R(dB) - 20\log d(km) - 20\log f(MHz) - 32.4$$

(10 dB, 10 dB)

$= 36.88 + 10\,dB + 10\,dB - 20\log 5 - 20\log 100 - 32.4$

$= 56.88\,dB - 32\,dB - 14\,dB - 40 = -27.12\,dBm$

$\approx -27\,dBm$

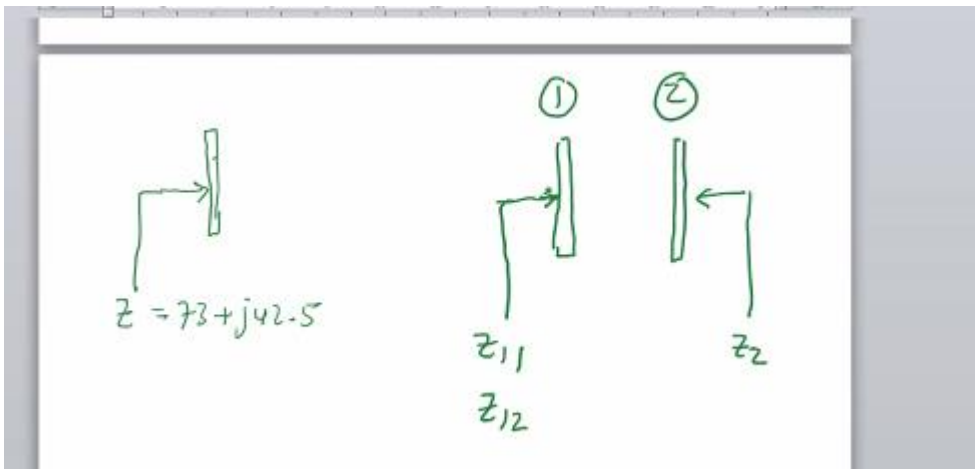
$= 2 \times 10^{-3}\,mW$.

$27\,dB = 500$

$0\,dBm = 1\,mW$

$-27\,dBm = \frac{1}{500} = 2 \times 10^{-3}\,W$

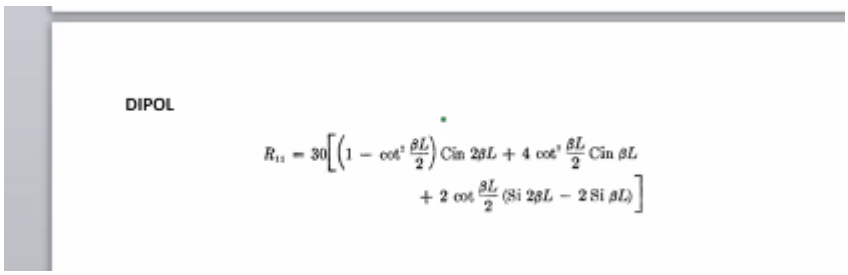
Antenna susun karena kita ingin mengarahkan sekehendak kita, bukan hanya meningkatkan gain tetapi membentuk pola radiasi.



Z12 karena imbas dari impedansi tetanganya







self-impedance

$$Z_{11} = R_{11} + jX_{11} = 30\,[\mathrm{Cin}\,(2\pi n) + j\,\mathrm{Si}\,(2\pi n)]$$

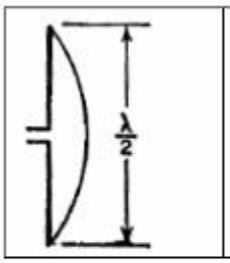
$$Z_{11} = 30\,[0.577 + \ln(2\pi n) - \mathrm{Ci}\,(2\pi n) + j\,\mathrm{Si}\,(2\pi n)]$$

self-resistance :

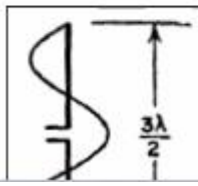
$$R_{11} = 30\,\mathrm{Cin}\,(2\pi n) = 30\,[0.577 + \ln(2\pi n) - \mathrm{Ci}\,(2\pi n)]$$

self-reactance

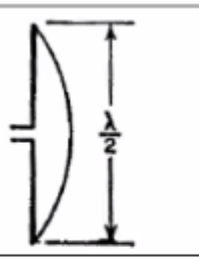
$$X_{11} = 30\,\mathrm{Si}\,(2\pi n)$$



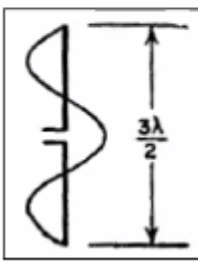
$R_{11} = 30\,\mathrm{Cin}\,(2\pi)$
$X_{11} = 30\,\mathrm{Si}\,(2\pi)$
$Z_{11} = R_{11} + jX_{11} = 73 + j42.5$ ohms



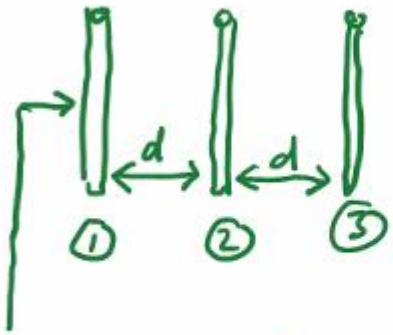
$Z_{11} = 30\,[\mathrm{Cin}\,(6\pi) + j\,\mathrm{Si}\,(6\pi)]$
$Z_{11} = 105.5 + j45.5$ ohms



$R_{11} = 30\,\mathrm{Cin}\,(2\pi)$
$X_{11} = 30\,\mathrm{Si}\,(2\pi)$
$Z_{11} = R_{11} + jX_{11} = 73 + j42.5$ ohms



$Z_{11} = 30\,[\mathrm{Cin}\,(6\pi) + j\,\mathrm{Si}\,(6\pi)]$
$Z_{11} = 105.5 + j45.5$ ohms



side-by-side

z_{11} z_{22} z_{33}

$z_{12}(d)$ $z_{21}(d)$ $z_{31}(2d)$

$z_{13}(2d)$ $z_{23}(d)$ $z_{32}(2d$

$= \left[-(-1-1)\right]\left[\int_0^{\frac{\pi}{2}} \frac{1}{2}\,d\theta - \frac{1}{2}\cdot\frac{1}{2}\int_0^{\pi}\cos 2\theta\, d2\theta\right]$ $d\phi = \frac{1}{2}\,d2\theta$

$= (2)\left[\frac{\pi}{2} - \frac{1}{4}\sin 2\theta\Big|_0^{\pi}\right] = (2)\left[\frac{\pi}{2} - 0\right] = \pi$ ✓

(a) $D_0 = \frac{4\pi U_m}{P_{rad}} = \frac{4\pi(1)}{\pi} = 4 = 6\ dB$ ✓

(b) $D_0 = \frac{41253}{(HP\text{-}1)(HP\text{-}2)}$

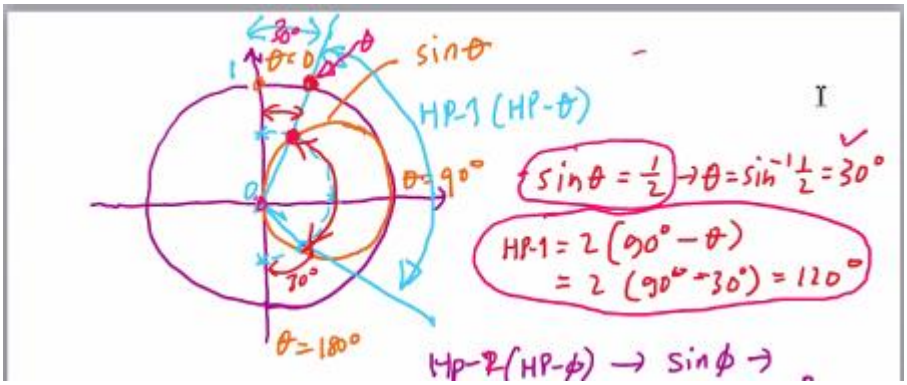
$U = \sin\theta \sin\phi \Rightarrow \frac{1}{2}$

$f(\theta,\phi) = \sin\theta \rightarrow \frac{1}{\sqrt{2}}$

$U = \sin\theta \sin\phi$

$\frac{1}{2} \in U = W/u^2 \in$ pola dunya

$\frac{1}{\sqrt{2}} \in f(\theta,\phi) \in$ pola medan



$\sin\theta = \frac{1}{2} \rightarrow \theta = \sin^{-1}\frac{1}{2} = 30°$ ✓

$HP\text{-}1 = 2(90° - \theta)$
$= 2(90° - 30°) = 120°$

HP-2 (HP-ϕ) → $\sin\phi$ →